Smart Inventory & Warehouse Management System

Project Overview

Smart Inventory & Warehouse Management System is a full-stack web application designed to efficiently manage inventory, suppliers, and stock movement. It simulates real-world warehouse operations and provides role-based access control for workers, admins, and super admins.

Objective

• Manage inventory and suppliers

• Track stock-in and stock-out operations

• Monitor low-stock items

• Generate reports using MongoDB aggregation

• Optimize query performance using indexing

System Requirements

Collections Used (MongoDB)

- items
- suppliers
- transactions
- users
- audit_logs
- item_requests

Core Functionalities

- Add and update stock
- Track incoming and outgoing goods
- Supplier management
- Transaction history tracking

- Role-based system access
- Item request workflow (approve/reject)

Features

➢ **Admin Dashboard**

- View analytics (stock, transactions, reports)
- Manage items and suppliers
- Review item requests

➢ **Worker Dashboard**

- View items and suppliers
- Perform stock IN/OUT
- Submit item requests

➢ **Data & Analytics**

- Low stock items detection
- Total stock value calculation
- Category distribution reports
- Movement summary reports
- Aggregation pipelines using MongoDB

➢ **Indexing (Important)**

Compound index used:

{ item_id: 1, date: 1 }

- Improves query performance
- Faster filtering and sorting
- Efficient transaction history queries

Tech Stack

❑ **Backend**

- Node.js (Express)

- MongoDB Atlas + Mongoose
- JWT Authentication
- Joi Validation
- Helmet, CORS, Rate Limiting

❑ **Frontend**

- React.js
- Tailwind CSS

❑ **Deployment**

- Vercel (Frontend)
- MongoDB Atlas (Database)

Architecture Overview

- Client Layer (Role-based dashboards)
- API Layer (REST APIs, middleware)
- Data Layer (MongoDB + Mongoose models)

Database Design

❑ **Items**

- name
- quantity
- price
- supplier_id

❑ **Suppliers**

- name
- contact
- address

❑ **Transactions**

- item_id
- type (IN / OUT)
- quantity
- date

Installation Guide

- git clone <your-repo>
- cd project
- npm install
- npm run dev

Deployment

- Live URL: https://udb-ca.vercel.app/login

Screens (Description)

- Login Page
- Dashboard
- Items Page
- Suppliers Page
- Transactions Page
- Reports Dashboard

Key Functionalities

- Inventory management
- Supplier management
- Aggregation queries
- MongoDB indexing
- Dashboard analytics
- Full-stack deployment

Future Improvements

- JWT Authentication enhancement
- Mobile responsiveness
- Advanced analytics dashboard
- AI-based stock prediction

Team Members

- Chayan Das – Backend & Integration
- Subhangkar Barui – Frontend Development
- Ritupan Daimary – Database & Indexing
- Mrinal Sahoo – Testing & Aggregation

Conclusion

This project successfully demonstrates a scalable and efficient inventory management system using MongoDB, with optimized performance through indexing and powerful analytics using aggregation.